

Wenlai Han

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RESEARCH INTERESTS	Reaction-network control and the differentiation of parallel reaction pathways; differentiable kinetic modeling and the development of automated reaction-discovery pipelines; skeletal and peripheral remodeling of C(sp ³)-rich compounds under dynamic kinetic control; and metal–ligand cooperativity.	
EDUCATION	Massachusetts Institute of Technology , Cambridge, MA Ph.D. Candidate, Organic Chemistry · <i>GPA: 5.00/5.00</i> XtalPi AI for Science Fellow; Advisor: Prof. Alison Wendlandt Carleton College , Northfield, MN B.A. in Chemistry and Computer Science ; minor in Music Performance. <i>Magna cum laude</i> with distinction in the chemistry major · <i>GPA: 3.92/4.00</i> Advisors: Prof. Matt Whited, Prof. David Alberg, and Prof. Jeff Ondich Stanford University , Stanford, CA High School Summer Visiting Student · <i>GPA: 3.93/4.00</i> China Agricultural University , Beijing, China Visiting High School Research Assistant (during academic breaks)	2023 – Present 2019 – 2023 2018 2016 – 2018
TECHNICAL SKILLS	Programming: Python, Julia, JavaScript, C, SQL. Scientific computing & modeling: Julia SciML ecosystem: symbolic reaction networks (Catalyst.jl), stiff ODE integration (DifferentialEquations.jl), adjoint sensitivity analysis and gradient-based optimization (SciMLSensitivity.jl), and derivative-free / global optimization (BlackBoxOptim.jl, JuMP.jl); differentiable kinetic modeling; rate-constant inference from time-resolved spectroscopic data. Software & ML: full-stack development, data-pipeline and API design, agentic LLM applications, and interactive data visualization.	
COMPUTATIONAL & SOFTWARE PROJECTS	• PocketForge – a full-stack design–make–test–analyze (DMTA) drug-design pipeline with validated data APIs and explainable selectivity scoring. • Know-de – an agentic LLM tutor that guides students through Socratic questioning rather than direct explanation. • Geothermal Data Web App – an interactive data dashboard built for Carleton Geology. • PM_{2.5} Sensor Pipeline – data processing and analysis for low-cost passive particulate sensors. • Natural Disaster Web App – an interactive interface visualizing global natural-disaster data.	
JOURNAL PUBLICATIONS	“Synthesis and Cooperative Reactivity of Sterically Encumbered Silyl and Silylene (^t BuP ₂ Si)Co Complexes”, Whited, M*; Pearson, C; Stewart, M; Lee-Zacheis, C; Khan, A; Han, W ; Giuliani, K; Kohen, D, <i>Organometallics</i> 2026, 45, 3, 297–305. [doi] “Migrating Group Strategy for Remote Functionalization of Seven-Membered Rings”, Han, W [†] ; Hwang, T [†] ; Lian, C; Kolb, S; Yamane, M; Palani, V; Wendlandt, A*, <i>J. Am. Chem. Soc.</i> 2025, 147, 35, 32077–32084. [doi] “Cobalt Silylenes as Platforms for Catalytic Nitrene-Group Transfer by Metal-Ligand	

Cooperation”, Whited, M*; **Han, W**; Jin-Lee, H; DiNardo, Z; Watson, E; Zhang, J; Kohen, D, *Angew. Chem. Int. Ed.* 2022, 61, e202205748. (Hot Paper) [doi]

“Novel Application of CO₂-Assisted High Pressure Processing in Cucumber Juice and Apple Juice”, Zhao, L; Qin, X; **Han, W**; Wu, X; Wang, Y; Hu, X; Ling, J; Liao, X, *LWT* 2018, 96, 491–498. [doi]

[†]These authors contributed equally.

SELECTED
CONFERENCES &
PRESENTATIONS

“Migrating Group Strategy for Remote Functionalization of Seven-Membered Rings”, **Han, W**; Hwang, T; Lian, C; Kolb, S; Yamane, M; Palani, V; Wendlandt, A, *Amgen Young Investigator Award Symposium* and *MIT–Merck Symposium*, 2025.

“Cobalt Silylenes as Platforms for Catalytic Nitrene-Group Transfer by Metal-Ligand Cooperation”, **Han, W**; Whited, M, *Gulf Coast Undergraduate Research Symposium*. Outstanding Presentation in Inorganic Chemistry Award and Travel Award, 2022.

“Metal-Ligand Cooperative Transfer of Alkyl Azide to Isocyanate Facilitated by a Si=Co Platform”, **Han, W**; Whited, M, 263rd *ACS National Meeting*. ACS–DIC Symposium Poster Prize and ACS–DIC Travel Award, 2022.

TEACHING
EXPERIENCE

Massachusetts Institute of Technology

- Teaching Facilitator, 5.12 Organic Chemistry I Fall 2023, Spring 2024
- Average student rating: 6.5/7

Carleton College

- Laboratory TA, Inorganic Chemistry Spring 2023
- Laboratory TA, Organic Chemistry I & II Fall 2020 – Fall 2022
- Tutor and Grader, Intermediate Chinese Fall 2020 – Fall 2022

LEADERSHIP &
OUTREACH

Massachusetts Institute of Technology

- EHS Representative and Safety Officer, Wendlandt Lab
- Volunteer, MIT Women+ in Chemistry (WIC+)
- Certified Mentor, Wendlandt Lab and MIT Department of Chemistry

SELECTED
AWARDS

Massachusetts Institute of Technology

- American Chemical Society, Division of Organic Chemistry, Travel Award
- Department of Chemistry [Mentorship Spotlight Award](#)

Carleton College

- [Jerry Mohrig Prize](#) in Chemistry
- Dean’s List
- [Travel Writing Contest](#), 3rd Place

Prior to Carleton College

- U.S. National Chemistry Olympiad, Gold Award (China Division)
- Future Business Leader Summit (MicroBiz International), 3rd Place — Team CEO, Business Simulation